The D. G. Ruparel College
OF ARTS, SCIENCE AND COMMERCE
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TITLE :

Aim:- To interface an R307/R305 optical finger-print sensor and a 1-channel relay module with a Raspberry Pi 4 and simulate a fingerprint-authenticated door-lock system.

* Hardware Requirements:-
1. Raspberry Pi Model A/B/B+
2. Jumper wires (Female to Male)
3. Relay Module
4. R307/R305 fingerprint sensor

* Software Requirements:-
1. Raspbian Stretch OS

1. Connect with Raspberry Pi's GPIO Pins.
- Relay Module

Device/Pin	Raspberry Pi 4	Physical Pin
Relay VCC	5V	2
Relay GND	GND	9
Relay IN	GPIO 17	11

- R307/R305 Fingerprint

R307S wire	R307S function	Raspberry Pin
• Red	VCC/5V	Pin 2
• Black	GND	Pin 6
• Yellow	TXD	Pin 10
• Green	RXD	Pin 8

* Software Installation

Step 1: Update Raspberry Pi packages

```
sudo apt update
sudo apt upgrade -y
```

Step 2: Install Python and serial support

```
sudo apt install -y python3 python3-pip python3-serial
```

Step 3: Install fingerprint library

```
pip3 install adafruit-circuitpython-fingerprint
```

Step 4: Enable Raspberry Pi UART

```
sudo raspi-config
```

Go to Interface Options --- Serial Port

Login shell over serial: No

Serial port hardware: Yes

```
sudo reboot
```

After reboot, check the serial device:

ls -l /dev/serial0

If Raspberry Pi OS restricts system-wide pip installation

```
python3 -m venv fingerprint-env
source fingerprint-env/bin/activate
pip install pyserial adafruit-circuitpython-
                    Fingerprint RPi.GPIO
```

*Code:-

•Fingerprint Enrollment

```
import serial
import time
import adafruit_fingerprint

uart = serial.Serial("/dev/serial0", baudrate=57600,
      timeout=1)
finger = adafruit_fingerprint.Adafruit_Fingerprint
                                (uart)

finger_id = int(input("Enter Fingerprint ID
                    (1-127): "))
```

```
print("Place finger on sensor...")
while finger.get_image() != adafruit_fingerprint.
                                OK:
  time.sleep(0.1)

print("Image captured")

if finger.image_2_tz(1) != adafruit_fingerprint.
                                OK:
  print("Image conversion failed")
  exit()

  print("Remove finger")
  time.sleep(2)

  print("Place the same finger again...")
  while finger.get_image() != adafruit_fingerprint.
                                OK:
    time.sleep(0.1)

  if finger.image_2_tz(2) != adafruit_fingerprint.
                                OK:
    print("Image conversion failed")
    exit()
```

```
if finger.store_model(finger_id) == adafruit_
                    fingerprint.OK:
    print("Fingerprint enrolled successfully")
    print("Stored ID:", finger_id)
else:
    print("Failed to store fingerprint")
```

- Main Door Lock Verification Program

```
import serial
import time
import RPi.GPIO as GPIO
import adafruit_fingerprint

#Change this GPIO if required
RELAY_PIN = 17

# Active-LOW Relay
RELAY_ON = GPIO.LOW
RELAY_OFF = GPIO.HIGH

GPIO.setmode(GPIO.BCM)
GPIO.setup(RELAY_PIN, GPIO.OUT, initial = RELAY_
                                    OFF)
```

```
try:
    uart = serial.Serial("/dev/serial0", baudrate = 57600,
                         timeout=1)
    finger = adafruit_fingerprint.Adafruit_Fingerprint
                         (uart)

    print("Fingerprint Door Lock System")
    print("Place finger")

    while True:
        result = finger.get_image()

        if result == adafruit_fingerprint.OK:
            print("Processing...")

            if finger.image_2_tz(1) != adafruit-
                                fingerprint.OK:
                print("Access Granted")
                GPIO.output(RELAY_PIN, RELAY_ON)

                print("Door Unlocked")
                time.sleep(5)

                GPIO.output(RELAY_PIN, RELAY_OFF)
                print("Door Locked")
```

```
else:
    print("Access Denied")
    GPIO.output(RELAY_PIN, RELAY_OFF)

    time.sleep(1)
    print("Place finger")

elif result != adafruit_fingerprint.NO_FINGER:
    print("Fingerprint sensor error")
    time.sleep(1)

except Exception as e:
    print("Communication error:", e)

finally:
    GPIO.output(RELAY_PIN, RELAY_OFF)
    GPIO.cleanup()
```

* Run the Main Program

```
python3 ~~entro~~ enroll.py
python3 door_lock.py
```